Bereich: Datenstrukturen (3)

Binärbaum	**Musterlösung**
Package: `de.dhbwka.java.exercise.collections.gas`	**Klasse:** `GasStations`

```java
package de.dhbwka.java.exercise.collections.gas;

public class GasPrices {

	private double diesel;
	private double superE5;
	private double superE10;

	public double getDiesel() {
		return diesel;
	}
	public void setDiesel(double diesel) {
		this.diesel = diesel;
	}
	public double getSuperE5() {
		return superE5;
	}
	public void setSuperE5(double superE5) {
		this.superE5 = superE5;
	}
	public double getSuperE10() {
		return superE10;
	}
	public void setSuperE10(double superE10) {
		this.superE10 = superE10;
	}

	@Override
	public String toString() {
		return "Diesel: " + this.diesel + ", SuperE5: " + this.superE5
			+ ", SuperE10: " + this.superE10;
	}
}
```

```java
package de.dhbwka.java.exercise.collections.gas;

import javax.swing.JButton;
import javax.swing.JFrame;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JPanel;
import javax.swing.JTextField;

import java.awt.BorderLayout;
import java.awt.GridLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.HashMap;
import java.util.Map;
import java.util.Map.Entry;
```

```java
@SuppressWarnings("serial")
public class GasStations extends JFrame implements ActionListener {

	private JTextField txtName = new JTextField();
	private JTextField txtDiesel = new JTextField();
	private JTextField txtSuperE5 = new JTextField();
	private JTextField txtSuperE10 = new JTextField();

	private final Map<String, GasPrices> stations = new HashMap<>();

	public GasStations() {
		super("Gas Stations");
		this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

		// inputs
		JPanel panInput = new JPanel(new GridLayout(4,2,5,5));
		panInput.add( new JLabel("Station Name") );
		panInput.add( this.txtName );

		panInput.add( new JLabel("Diesel") );
		panInput.add( this.txtDiesel );

		panInput.add( new JLabel("Super E5") );
		panInput.add( this.txtSuperE5 );

		panInput.add( new JLabel("Super E10") );
		panInput.add( this.txtSuperE10 );

		// save button
		JPanel panButtons = new JPanel();
		JButton btnSave = new JButton("Save");
		JButton btnShow = new JButton("Show all");

		// example with single action listener and action commands
		btnSave.addActionListener(this);
		btnShow.addActionListener(this);
		btnSave.setActionCommand("save");
		btnShow.setActionCommand("show");

		panButtons.add(btnSave);
		panButtons.add(btnShow);

		this.add(panInput); // implicit: BorderLayout.CENTER
		this.add(panButtons, BorderLayout.SOUTH);


		this.pack();
		this.setVisible(true);
	}

	public static void main(String[] args) {
		new GasStations();
	}

	@Override
	public void actionPerformed(ActionEvent e) {
		switch (e.getActionCommand()) {
```

```
            case "save":
                this.saveCurrentInput();
                break;
            case "show":
                this.showAllStations();
                break;
            default:
                break;
            }
    }

    private void saveCurrentInput() {
            String station = this.txtName.getText();
            if ( station.isEmpty() ) {
                JOptionPane.showMessageDialog(this,
                    "Please provide a station name");
                return;
            }
            GasPrices gp = new GasPrices();
            gp.setDiesel(this.getValue(this.txtDiesel));
            gp.setSuperE5(this.getValue(this.txtSuperE5));
            gp.setSuperE10(this.getValue(this.txtSuperE10));

            this.txtName.setText("");
            this.txtDiesel.setText("");
            this.txtSuperE5.setText("");
            this.txtSuperE10.setText("");

            this.stations.put(station, gp);

            JOptionPane.showMessageDialog(this, "Saved: " + station
                + " (" + gp + ")");
    }

    private double getValue(JTextField txt) {
            String txtVal = txt.getText();
            if ( !txtVal.isEmpty() ) {
                txtVal = txtVal.replace(',', '.'); // optional comma support
            }
            try {
                return Double.parseDouble(txtVal);
            } catch (Exception e) {
                return -1;
            }
    }

    private void showAllStations() {
            StringBuilder sb = new StringBuilder();
            for( Entry<String, GasPrices> e : this.stations.entrySet() ) {
                sb.append(e.getKey() + ": " + e.getValue() + "\n");
            }
            JOptionPane.showMessageDialog(this, sb.toString().trim());
    }
}
```